

Historical Review of BCI Assistive Technology in Musical Performance

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Abstract

Brain-computer interfaces (BCIs) have been used as both therapeutic and artistic tools for musical performance by individuals with restrictive mobility, including people with ALS and other disabilities. Yet BCIs were not originally developed with assistive or accessibility goals in mind. Rather, they have undergone a gradual evolution determined by persistent technical and design challenges. This brief historical overview will examine the development of brain-computer musical interfaces (BCMIs) over multiple decades, along with the challenges of each era. Some challenges include brainwave instability, training burden, limited expressive control, and design approaches that have unintentionally excluded users. Designers later included multimodal input systems, integrated systems with assistive technologies, and adopted accessibility-focused approaches to support broader inclusiveness in BCMI technology.

ACM Reference Format:

Kimberly By Goytia. 2025. Historical Review of BCI Assistive Technology in Musical Performance. In . ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Personal Motivation

Music and musical performance have always been my primary outlets during periods of personal struggle. From middle school choir to present-day college a cappella, music has always given me a sense of community, no matter our differences. I was always able to engage with people from so many different walks of life in musical spaces who I would never have otherwise met. For me and so many others, music was never just about hitting the right note. It was about connection, belonging, and collaboration. This understanding of musical performance as a space for community building shaped my interest in researching the history of BCMIs. Reading, Brain Body Jockey also deepened this interest as I read about how BCI helped a video jockey with ALS, a way to physically engage the audience through the movement of stage robotic arms. His performance fostered a sense of unity among the crowd that he had not experienced before. Musical performance is a collective experience, and creating accessible ways to foster that through BCMIs is absolutely beautiful.

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Conference'17, Washington, DC, USA

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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

2 Introduction

Brain-computer interfaces (BCIs) have gained popularity across the research and design spaces as both therapeutic and artistic tools for musical performance by people with restricted mobility. These communities often include those with ALS and other physical disabilities like quadriplegia. Although BCIs now have a reputation as assistive technologies, they did not initially have disabled users in mind. Instead, the field has evolved through a series of technical design challenges that have shaped BCI into an access-oriented technology. In an effort to uncover the history of BCMIs (brain-computer musical interfaces), this paper will trace the linear evolution of the field from early cybernetics, through the formalization of BCMI as a field, to assistive BCIs for ALS, hands-free, accessible digital musical instruments, and mixed-reality BCI integrations.

3 Historical Evaluation of BCMIs

3.1 Early Cybernetics (1960s-1990s)

An early EEG (brainwaves) and biofeedback (EMG electric muscle signals) music experiment explored whether internal physiological state could potentially influence basic sound production. At this stage, many of the signals from participants' brain waves were not fully detectable by the EEG machines. To obtain a strong enough trace for the system to produce a sound, participants would need to concentrate intensely. Pressing's paper on the cybernetic analysis of interactive performance systems identified additional problems with early EEGs: they were often disrupted by environmental and signal noise, placed a high cognitive load on participants, and mapped brainwave responses to the wrong sound outputs [8]. Real-time interactive systems were found to be quite unstable due to noise conflicts and variability in user signals. At this point in time, disabled performers were not yet considered part of the co-design process, so many of the challenges faced by designers were based on able-bodied-centric assumptions.

3.2 Birth of BCMI Field (1990s-2000s)

In the late 1990s, researchers such as Eduardo Reck Miranda helped formalize BCMI as a new and unique field focused on expressive interaction [6]. While this period introduced more structured EEG-to-sound mappings, many challenges remained unaddressed. EEG signals remained highly sensitive to participants' emotional states due to similar fatigue-related concentration issues. This once again led to inconsistent musical output. The Plymouth BCMI project also found that motor imagery classifiers drifted over time, leading to unreliable performance with the BCMI [5]. At this stage of the evolution of BCMI, disabled users were occasionally mentioned

within the research as potential candidates or participants for using BCMI, but they were not regularly cited as potential co-designers.

3.3 Assistive BCIs (2000s-2020s)

Beginning in the early 2000s, there was a shift in the BCI field toward more formally supporting and centering disabled users in research. More specifically, there was a rise in research applications of BCI for individuals with ALS and locked-in syndrome, which are both conditions that affect the nervous and muscular systems. There was still a desire in this era to make BCI a creative assistive tool for these two communities, so projects such as Brain Painting provided users, particularly those with locked-in syndrome, with a creative outlet. Although fatigue and performance issues were improving, a new issue emerged related to facilitators' overreliance on calibrating the machines [4]. This dependence, along with the short time a user could actually use the device, posed further limitations on the technology's future as an assistive creative tool. By the 2010s, researchers began to dive into multimodal input methods by using EEG brain waves with a combination of eye-tracking and EMG signals with the aim of reducing user fatigue, improving machine reliability, and increasing the usability time. However, musical accessibility remained a core creative issue, with a survey finding that fewer than five percent of the accessible musical instruments at the time were reliable for users with limited limb movement [3]. This jump-started a new phase of BCMIs that embraced a wide range of hands-free controls, such as gaze, head movement, EEG, EMG, and facial [2]. Now, with an increase in multimodal BCMI research, there is a deeper emphasis on addressing issues such as fatigue in relation to prior BCMI work with disabled communities.

3.4 Mixed Reality and Hybrid Performance (2020s-Present)

After the introduction of multimodal design inputs in BCMI, which have helped reduce user fatigue and improve usability, a new frontier has been reached. BCMI has increased in use within the immersive performance space, which combines multimodal inputs, mixed reality, and robots. Projects like BCGJam and the embedding of EEG-based controls into mixed reality spaces have enabled performers to manipulate sound and visuals in new ways [9]. Additionally, SSVEP-based (Steady-State Visually Evoked Potential) BCIs have improved signal detection, enabling smoother signal tracing to sound with quicker training times [10]. Although mental fatigue and reliance on the facilitator for calibration have vastly improved, the new challenge with visual models is vision fatigue, especially when they rely on gaze triggers. The most recent development in this era of the field and the introductory paper to this research era for me is the Brain-Body Jockey project, which extends the hybrid multimodal approaches into robotics. This project's system focused on co-designing with a video DJ with ALS to create a more engaging and community-centered performance experience. Their system used the DJ's brain waves to control the movement of robotic arms on stage, whilst his eyes controlled his music set. This transformed his performance into a more interactive experience with the audience, and those who attended later said they felt more connected to him through the use of the robotic arms, which signaled different crowd movements [1].

4 First-Hand Account: Contemporary BCMI Performance (2015)

For a first-hand account of BCMI in action, I observed a BCMI-based musical performance from the short documentary of the Activating Memory ensemble performance [7]. Four participants with either locked-in syndrome or severe motor impairment due to an accident used EEG and eye-gaze-based interfaces to perform a string quartet together, both competitively and collaboratively. Based on which bright light the users looked at while a certain musical phrase was played, the machine would detect it and send it to the violinists. Many of these participants had prior musical experience before becoming physically impaired, so being able to conduct music again while collaborating with other disabled musicians made many of them happy. Overall, this performance highlights how meaningful creative assistive technologies can be for communities with limited mobility, not only providing an outlet but also enabling them to engage with a musical community again.

5 Personal Reflections on Disability and BCI

My engagement with disability and BCI assistive technology was very limited before I entered this course. None of my family members or close friends has relied on BCIs, and my early understanding of the technology was largely shaped by media like the public awareness campaign for ALS via the Ice Bucket Challenge. Through that campaign, I was able to see BCI and eye-tracking technology in action for the first time among people with ALS. Engaging with BCMI literature shaped my understanding of accessibility technologies as an examination of how design assumptions shape the technologies disabled communities have access to. This new understanding of assistive technology helped reshape my understanding of design exclusivity and our responsibility as technologists to continue advocating for co-design with the communities we are building for.

6 Disability Solidarity in Academic Spaces

While assessing how disability solidarity could look in music classes at Smith in relation to BCMIs, I immediately thought of MUS238: Laptop Orchestra. This course, offered in spring 2025 at Smith College, centered on computers as the primary performance instrument. This class presents a ripe opportunity to integrate access-oriented interfaces, such as BCMIs, for further creative exploration, including disabled student musicians.

7 Conclusion and Future Directions

While BCMIs have made meaningful progress toward supporting musical performance for people with restricted physical disabilities, there are still a few remaining challenges. To address these issues, future research must focus on co-designing with disabled artists so that challenges like signal instability, hardware accessibility, and fatigue are no longer present. Overall, BCMIs hold the potential to increase access to musical performance and reinvent what performance could look like in the future for disabled artists.

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